

Class Project

Proposal Due May 17, 2026 at 11:59pm

1 Objective

The purpose of the project is to apply data-parallelism and CUDA features to a more substantial piece of code than in the lab assignments. This could take many forms, including:

- Thorough performance analysis and improvement of existing GPU code or architectures
- Implement a GPU version of some existing computationally-intensive CPU code
- Reproduce some existing GPU research work
- Do novel GPU research
- And more!

Ultimately, the goal is to demonstrate a strong understanding of GPU programming and data-parallel computing concepts through an open-ended project. Please refer to the example project ideas provided on the course website for guidance.

2 Evaluation Criteria

Your project will be evaluated based on the following criteria:

- Demonstration of functionality and performance (50%): Produce correct results with good speedup relative to base code and/or existing implementation
- Final presentation (20%): Quality of oral delivery
- Report (20%): Clear, well-written, and organized report with in-depth analysis of the problem, the approach, and the results
- Code quality (10%): well-documented and organized code

3 Project Milestones

3.1 Project proposal: May 17 Sunday, 11:59pm

Short project proposal (2-page max) should be submitted online. This proposal should include:

- a short description of the algorithm to be parallelized in CUDA
- a list of references
- a short summary of your parallelization and optimization strategies and/or how you want to improve on existing versions
- a break-down of the work into a short list of tasks with tentative completion dates

3.2 Final presentation: June 9 Tuesday, in class

You will give a 5-minute talk about your project during the class hours. The talk should include a brief introduction (including motivation and background) for your work, key design ideas and implementation details, experimental results, and conclusion. The slides and the talk can be in either English or Korean.

3.3 Final report: June 14 Sunday

Your final project report should be submitted online. It should be a maximum seven pages (10pt font, single-column article format), not including References and Appendix.

1. The problem statement and motivation (1/2 page)
2. Brief summary of existing work in the literature, with citations (1/2 page)
3. Overview of your parallel algorithm design (2 page):
 - (a) Detailed design and implementation description with pseudo-code
 - (b) Short examples that show how your code works in different scenarios. Choose the example carefully to highlight the major technical capabilities and limitation.
4. Evaluation (2-4 pages):
 - (a) Evaluation methodology (system configurations, input dataset, compilation/runtime parameters, etc.)
 - (b) Experimental results: Compare the performance of your algorithm with a baseline sequential version and/or other parallel versions
5. References
6. Appendix: a description of where to find your source code, executable, and input programs, and how to run the executable for example inputs